**Example 2.3**

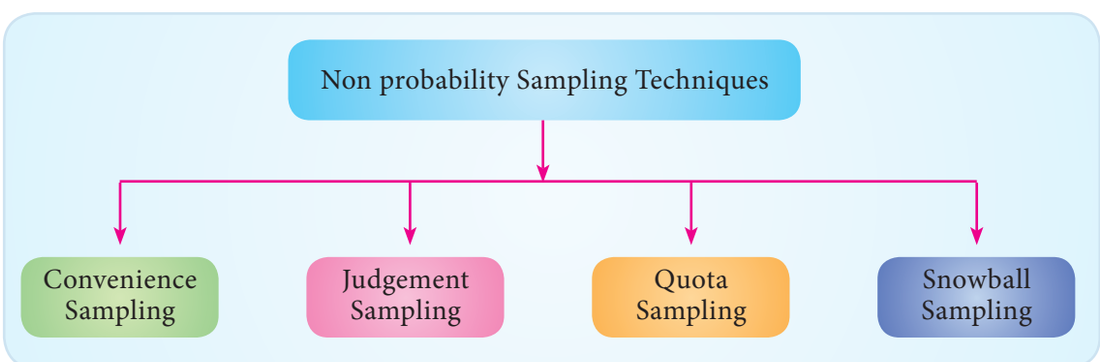
Suppose a systematic random sample of size $n = 10$ is needed from a population of size $N = 200$, the sampling interval $k = \frac{N}{n} = \frac{200}{10} = 20$. The first sampling interval consists of numbers 1 to 20. If the randomly selected number (random starter) is 7, the systematic sample will consist units corresponding to positions 7, 27,47,67,87,107,127,147,167,187.

## Applications:

- Systematic sampling is preferably used when the information is to be collected from trees in a highway, houses in blocks, etc,.
- This method is often used in industry, where an item is selected for testing from a production line (say, every fifteenth item in the order of production) to ensure that equipments are working satisfactorily.
- This technique could also be used in a sample survey for interviewing people. A market researcher might select every 10th person who enters a particular store, after selecting a person at random as a random start.

## 2.8 Non-probability sampling:

Non probability sampling is the sampling procedure in which samples are selected based on the subjective judgment of the researcher, rather than random selection. This is used when the representativeness of the population is not the prime issue. Convenience or judgments of the investigators play an important role in selecting the samples. In general, there are four types of non probability sampling called convenience sampling, judgment sampling, quota sampling and snowball sampling.




### 2.8.1 Convenience Sampling:

The samples are drawn at the convenience of the investigator. The investigator pick up cases which are easily available units keeping the objectives in mind for the study.



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